

FRAUD DETECTOR



Architecture & Model Design

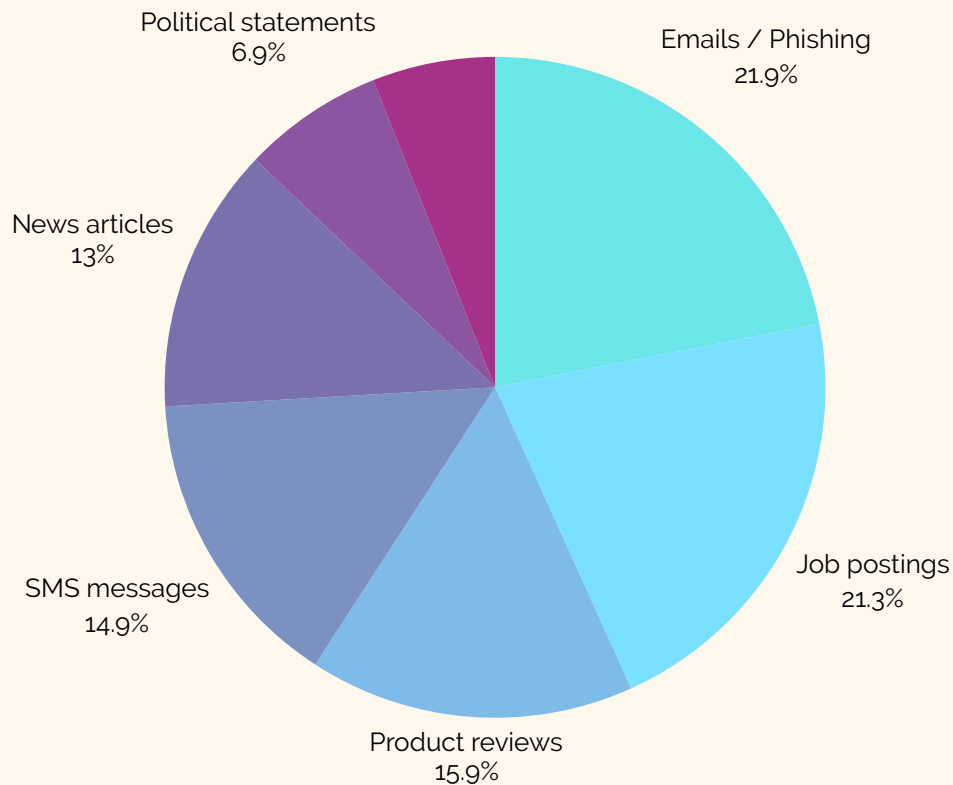
Made By

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Project Overview

The platform is a domain-independent fraud detection system for multiple textual sources:

- Emails / Phishing
- Job postings
- Product reviews
- SMS messages
- News articles
- Political statements
- Twitter rumors



It provides fraud probability (0–100%) and highlights suspicious text to the end-user.

The platform includes:

- FastAPI backend with domain-specific routes
- Frontend for copy-paste detection, news/job feeds, and tweet links
- LoRA-adapted DeBERTa-v3-base models for efficient multi-domain detection
- Web extension for real-time email scanning
- Optional N8N integration for automated pipelines and alerts

Dataset Overview



We used the DIFrauD dataset de hugging face

95,854

Total samples

100 %

29 %

37,282

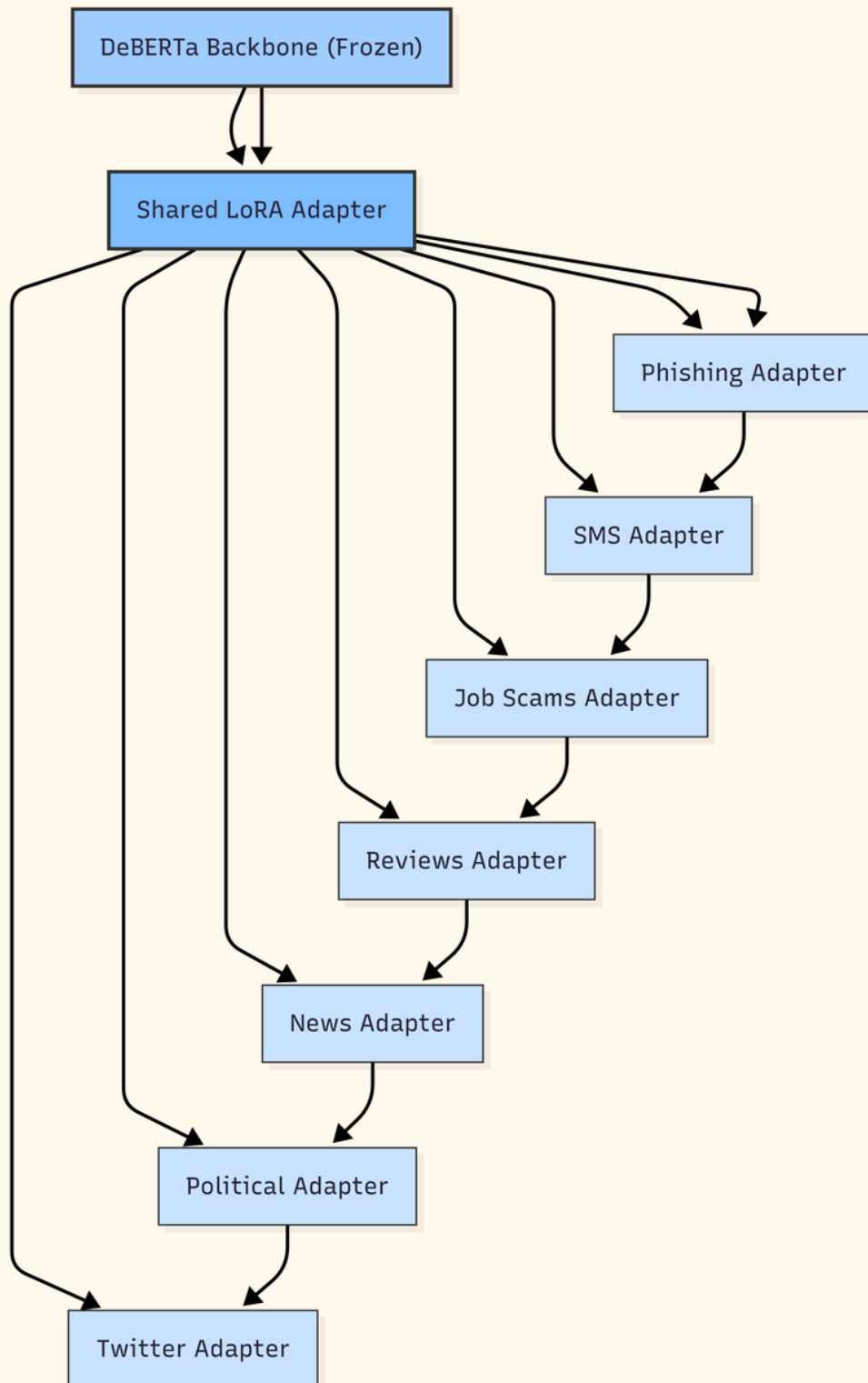
Deceptive

58,572

Non-deceptive

71 %

Model Architecture



Model Architecture

Backbone:

DeBERTa-v3-base

- retrained transformer encoder
- Frozen during training to retain general language understanding
- Handles both short texts (SMS, tweets) and long texts (news, reviews)
- Captures semantic, syntactic, and tone cues

Shared

LoRA Adapter

- Trains on combined data from all domains
- Learns patterns like: urgency, exaggerated claims, authority manipulation, suspicious links
- Lightweight (~30–50MB) and modular
- Reduces domain duplication

Domain

Specific LoRA Adapters

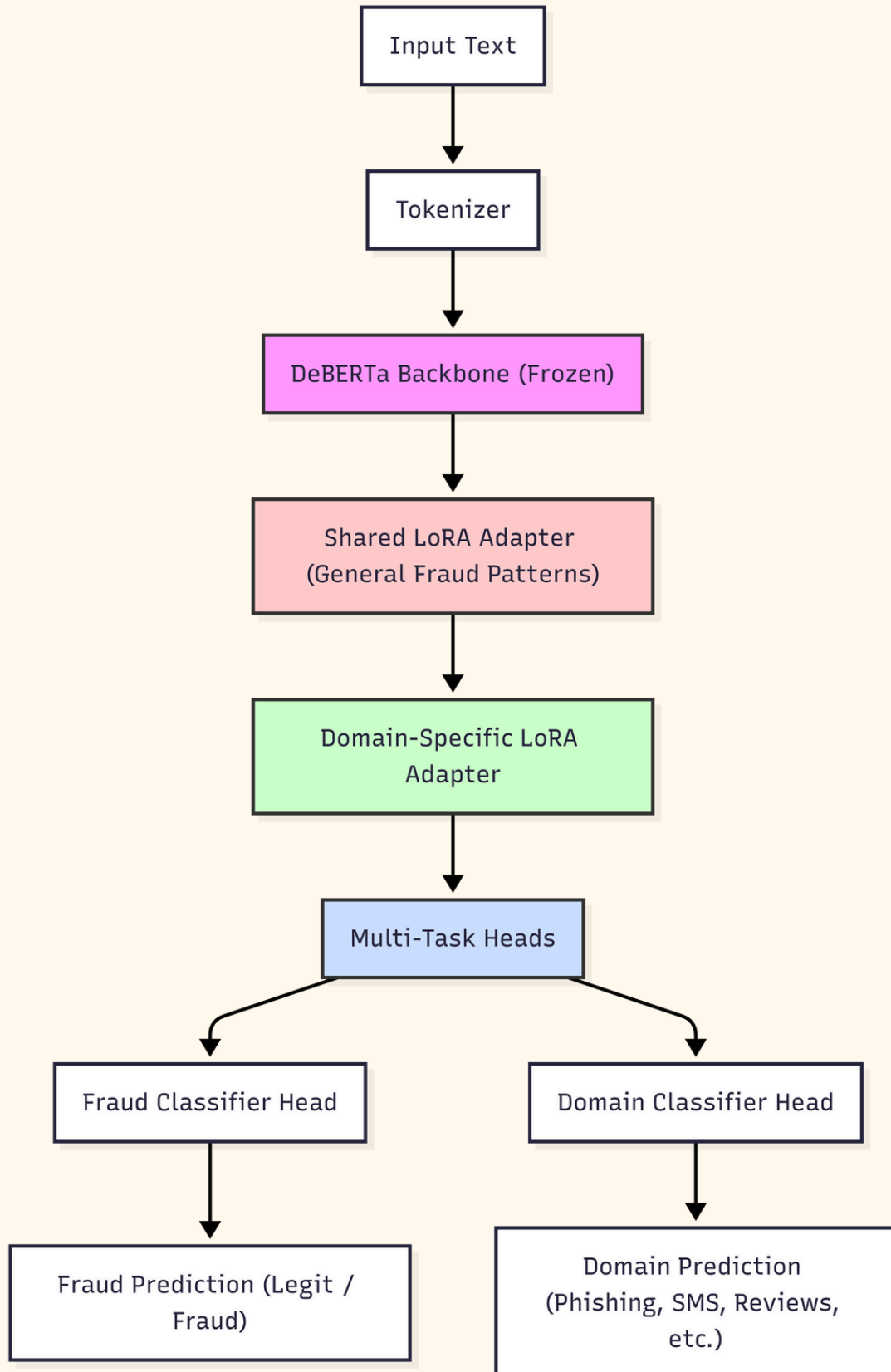
trains lightweight domain-specific LoRA adapters on top of the frozen shared adapter

Explainability

with SHAP or LIME

.Integrating explainability techniques like SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-agnostic Explanations)

Workflow



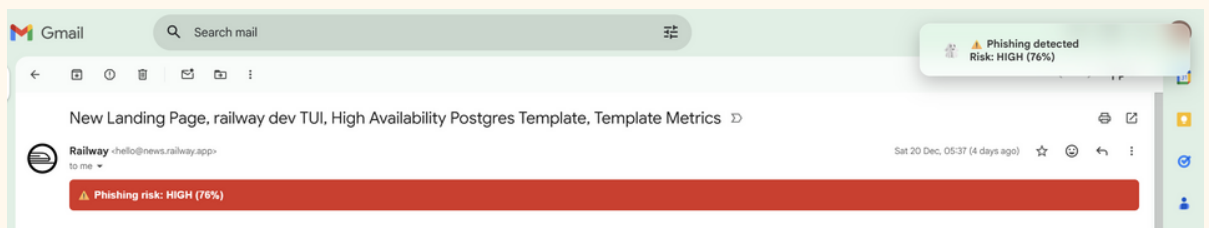
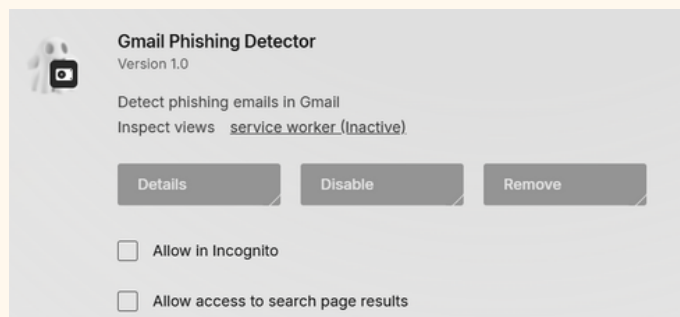
API & Frontend Architecture

Frontend Interface

Developed a user-friendly web application allowing users to interact with the fraud detection system. Features include copy-paste text input, job feed, news feed, and tweet analysis for real-time fraud detection.

Web Extension

Created a Chrome extension for email fraud detection, providing users with real-time alerts and highlighting suspicious content directly in their inbox.



API & Frontend Architecture

Automation with OPAL

Integrated workflow automation using OPAL to streamline tasks and detection fraud

API Integration for Developers

Implemented a fully documented FastAPI REST API, enabling developers to integrate the fraud detection models into external applications or services.

Opal

